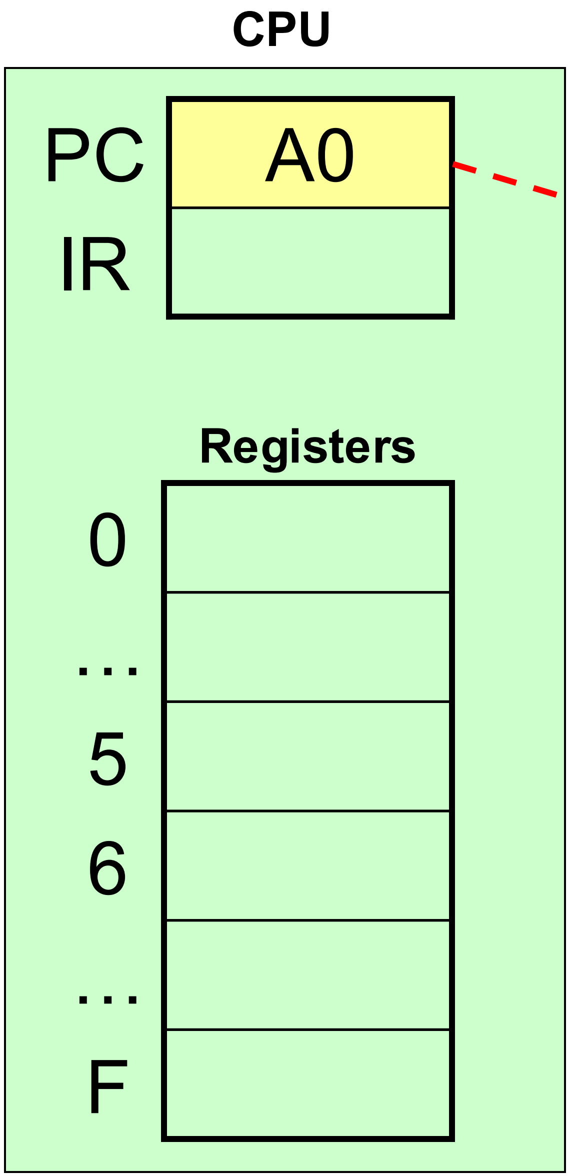
High level-Machine language relationship

C++:

TotalCost = Price + ShippingCharge;

Assembly:	**Machine Code (Hex):**
LD R5, Price	156C
LD R6, ShippingCharge	166D
ADDI R0, R5 R6	5056
ST R0, TotalCost	306E
HLT	C000

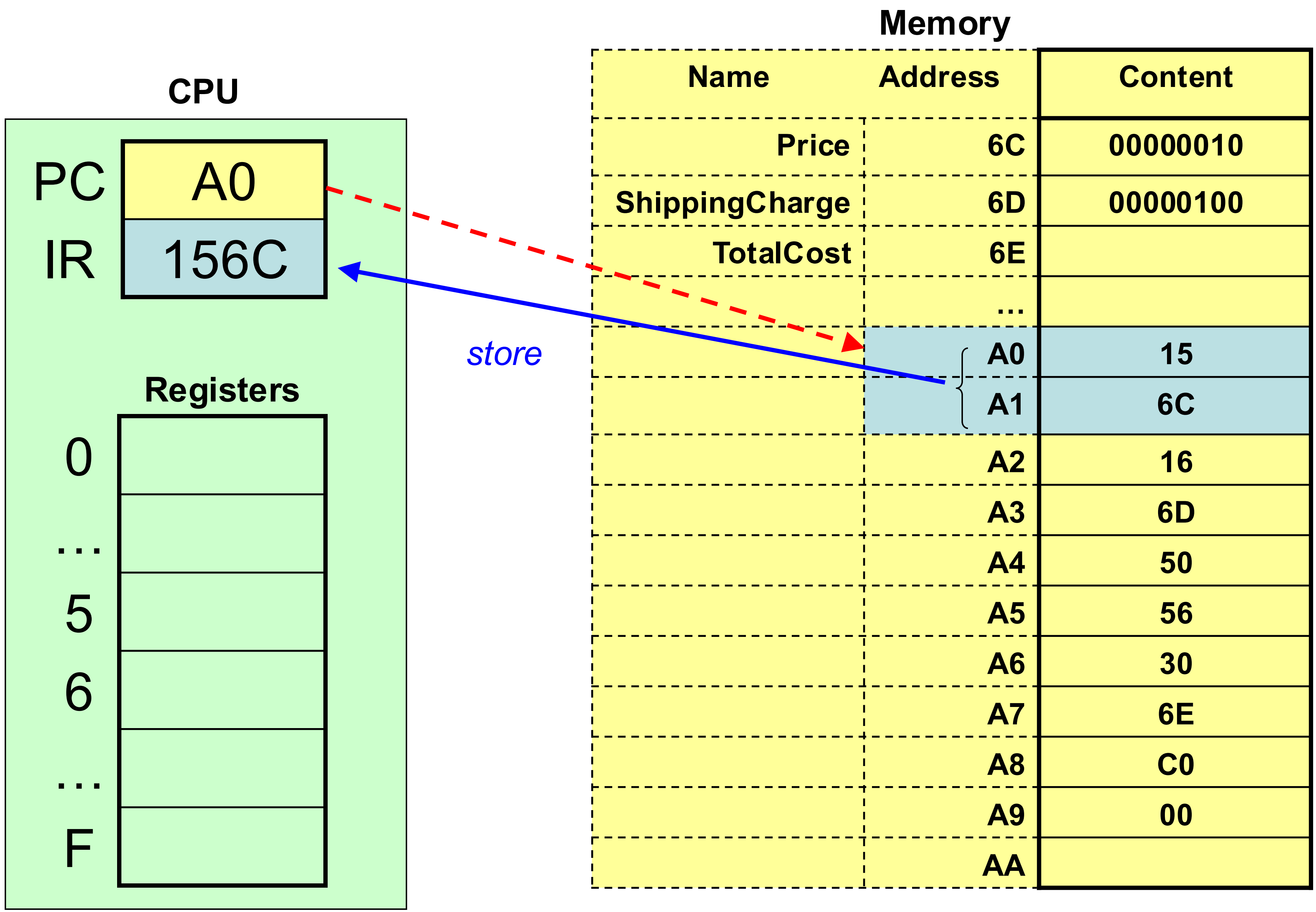
Start executing instructions



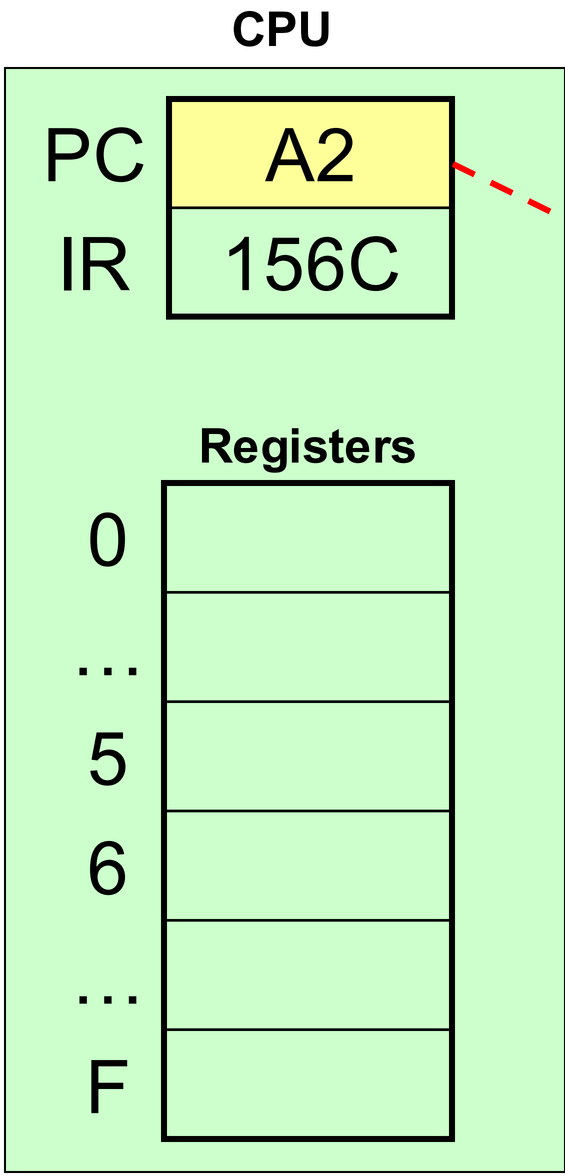
Memory

Name	Address	Content
Price	6C	00000010
ShippingCharge	6D	00000100
TotalCost	6E	
	...	
	A0	15
	A1	6C
	A2	16
	A3	6D
	A4	50
	A5	56
	A6	30
	A7	6E
	A8	C0
	A9	00
	AA	

a) **Fetch -** instruction from A0 and A1 – 15, 6C and store it in IR; increment PC to A2



a) **Fetch -** instruction from A0 and A1 – 15, 6C and store it in IR; increment PC to A2



Memory

Name	Address	Content
Price	6C	00000010
ShippingCharge	6D	00000100
TotalCost	6E	
	…	
	A0	15
	A1	6C
	A2	16
	A3	6D
	A4	50
	A5	56
	A6	30
	A7	6E
	A8	C0
	A9	00
	AA	

b) **Decode** – load register 5 with value stored at **Price** (memory location 6C)

CPU

PC	A2
IR	156C

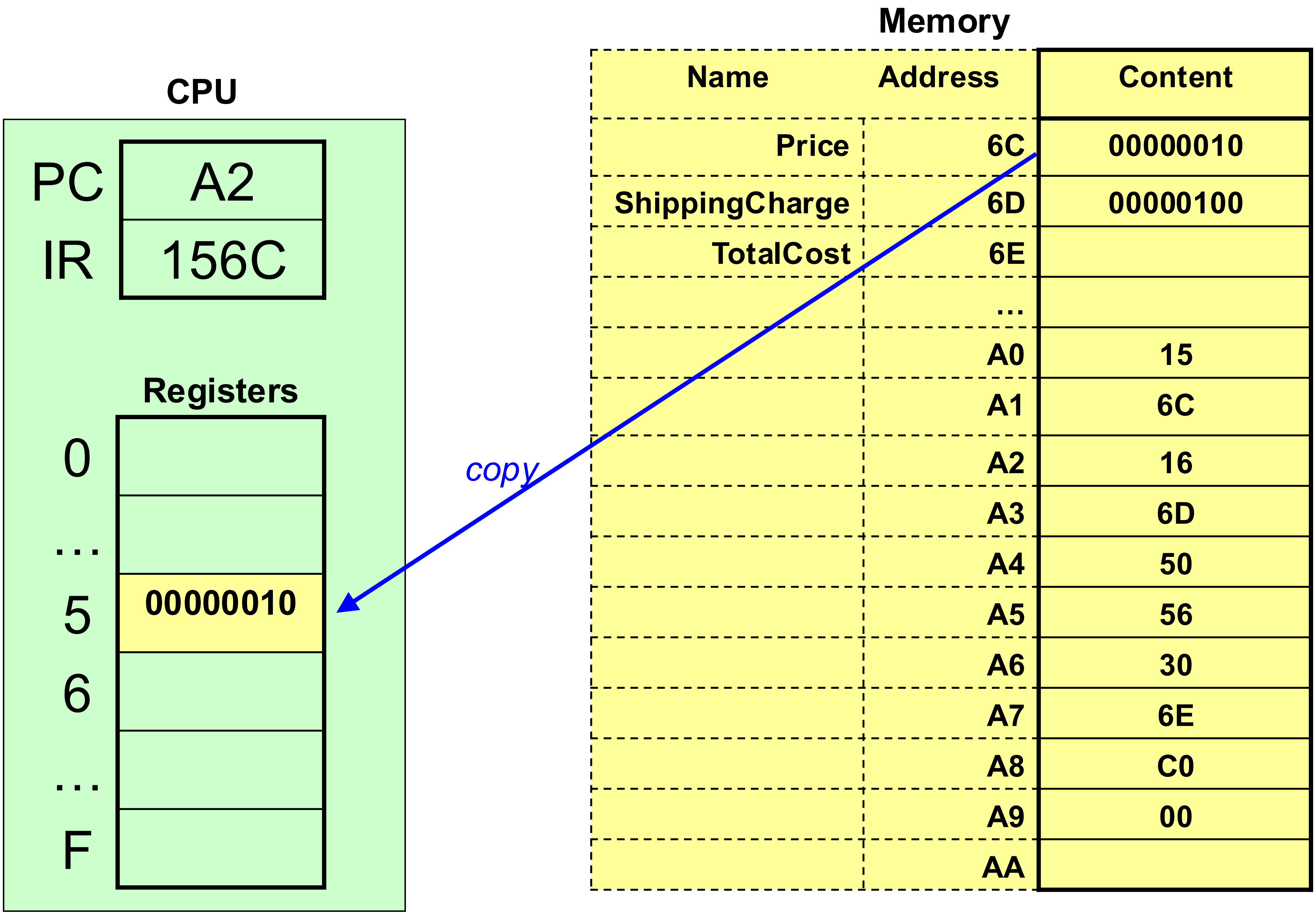
Registers

0	
…	
5	
6	
…	
F	

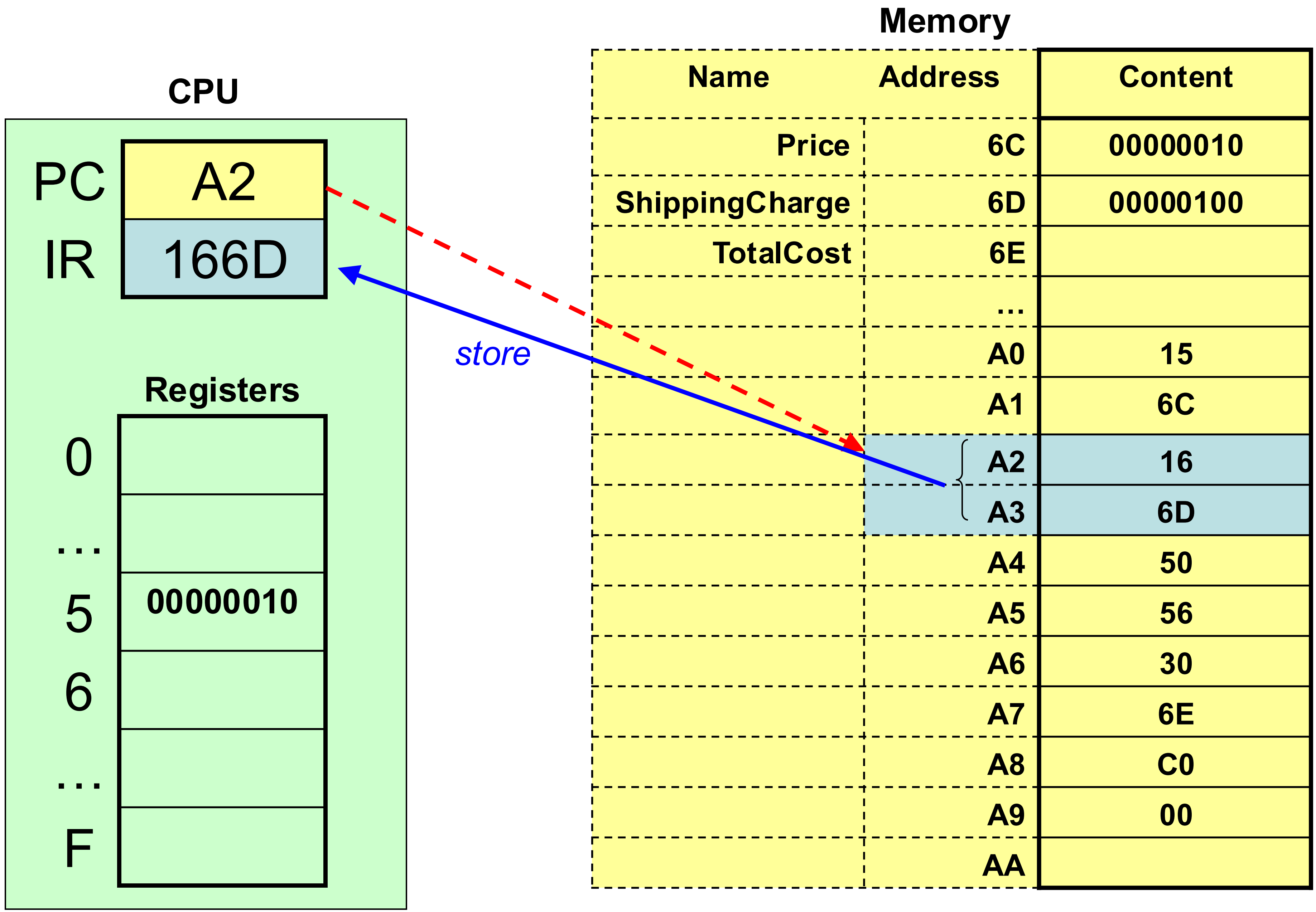
Memory

Name	Address	Content
Price	6C	00000010
ShippingCharge	6D	00000100
TotalCost	6E	
	…	
	A0	15
	A1	6C
	A2	16
	A3	6D
	A4	50
	A5	56
	A6	30
	A7	6E
	A8	C0
	A9	00
	AA	

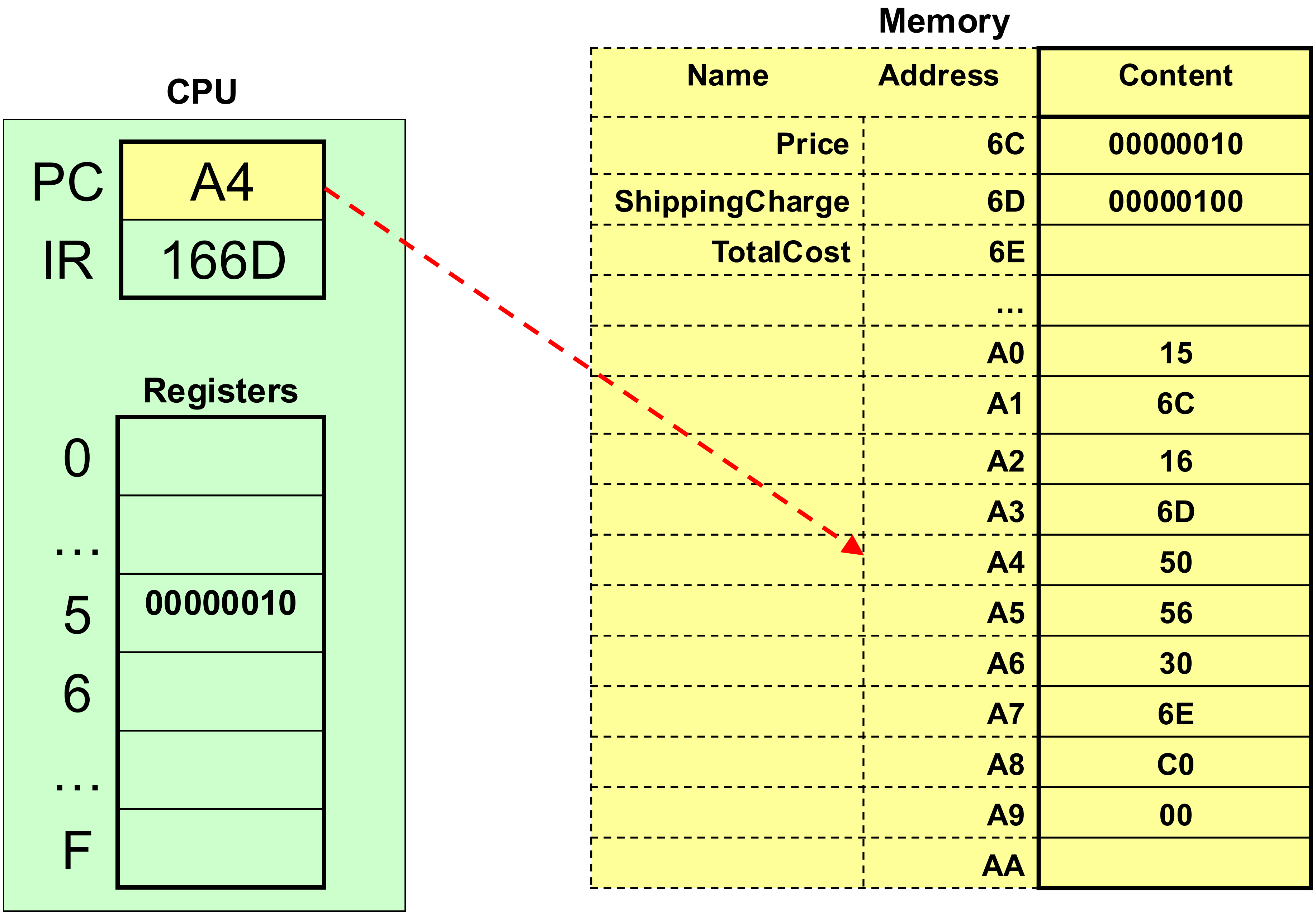
c) **Execute** – copy value from memory location 6C to register 5



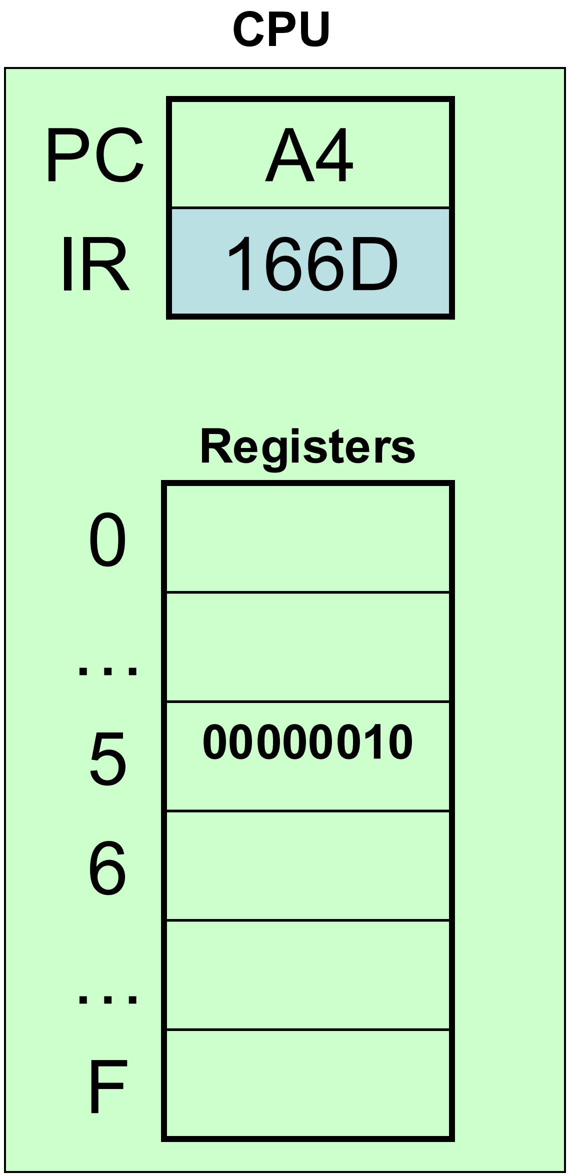
a) **Fetch -** instruction at A2 and A3 – 16, 6D and store it in IR; increment PC to A4



a) **Fetch -** instruction at A2 and A3 – 16, 6D and store it in IR; increment PC to A4



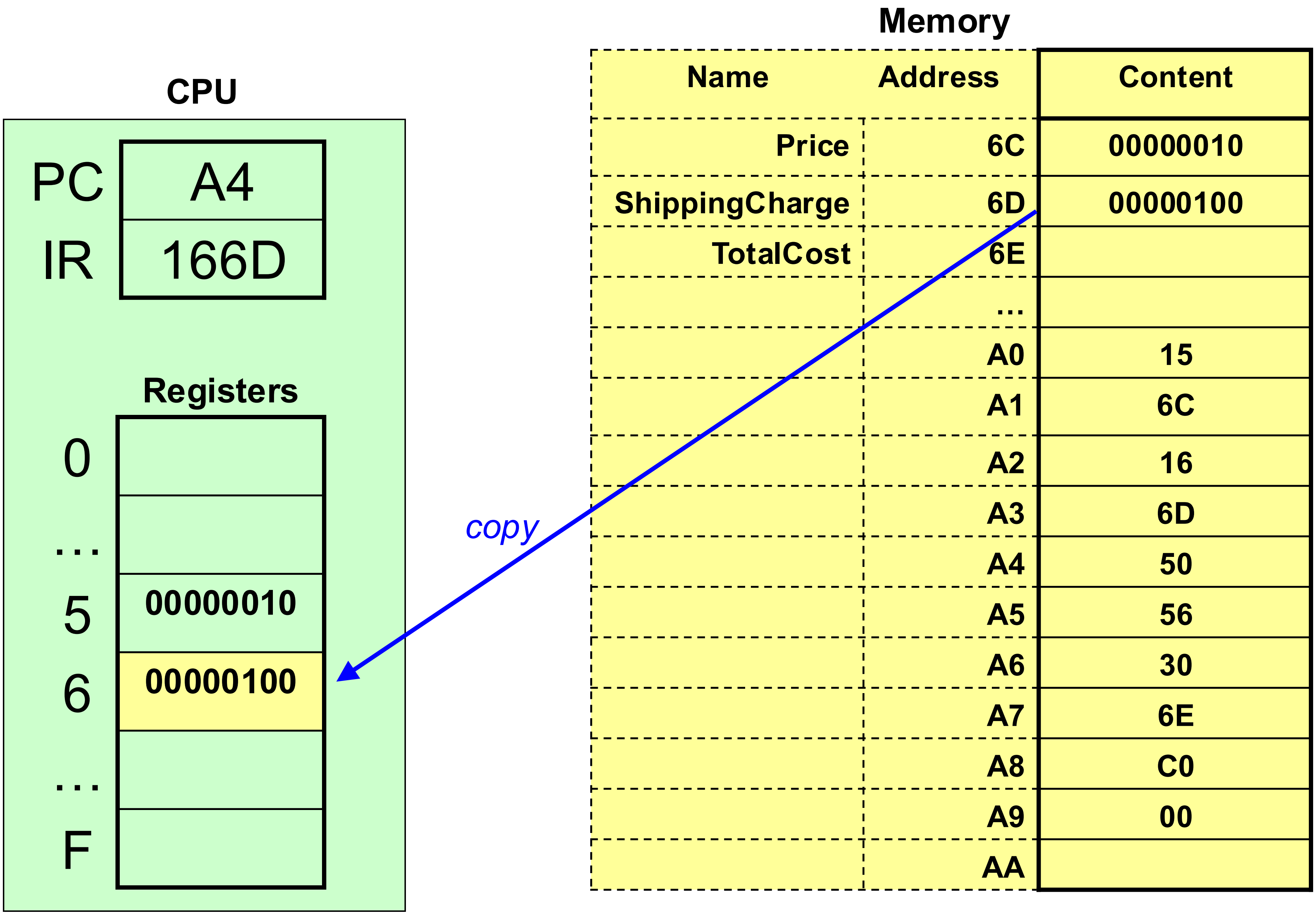
b) **Decode** – load register 6 with value stored at **ShippingCharge** (memory location 6D)



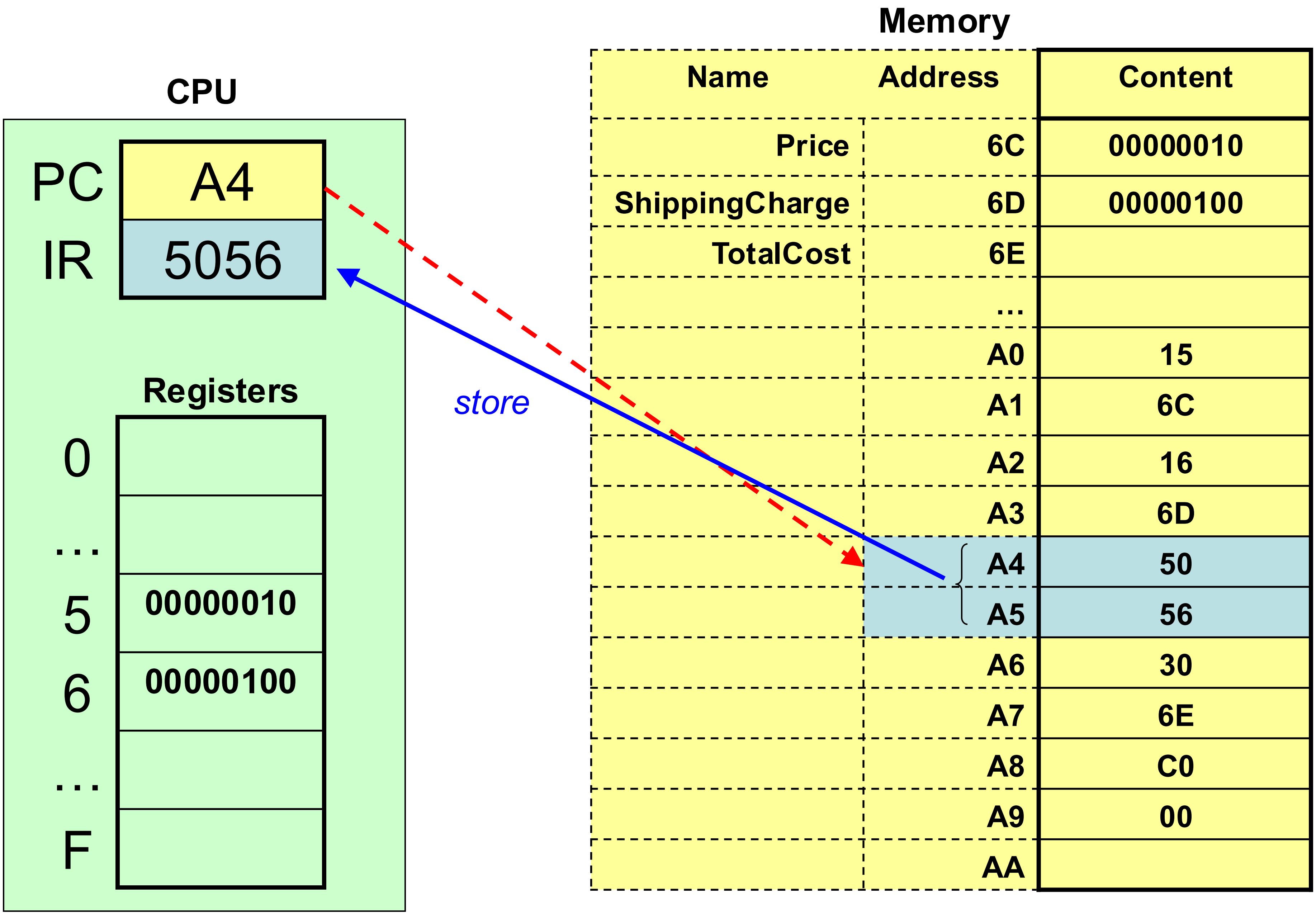
Memory

Name	Address	Content
Price	6C	00000010
ShippingCharge	6D	00000100
TotalCost	6E	
	…	
	A0	15
	A1	6C
	A2	16
	A3	6D
	A4	50
	A5	56
	A6	30
	A7	6E
	A8	C0
	A9	00
	AA	

c) **Execute** – copy contents of 6D to register 6



a) **Fetch -** instruction at A4 and A5 – 50, 56 and store it in IR; increment PC to A6



a) **Fetch -** instruction at A4 and A5 – 50, 56 and store it in IR; increment PC to A6

CPU

PC	A6
IR	5056

Registers

0	
…	
5	00000010
6	00000100
…	
F	

Memory

Name	Address	Content
Price	6C	00000010
ShippingCharge	6D	00000100
TotalCost	6E	
	…	
	A0	15
	A1	6C
	A2	16
	A3	6D
	A4	50
	A5	56
	A6	30
	A7	6E
	A8	C0
	A9	00
	AA	

b) **Decode** – add values in registers 5 and 6, store result in register 0

CPU

PC	A6
IR	5056

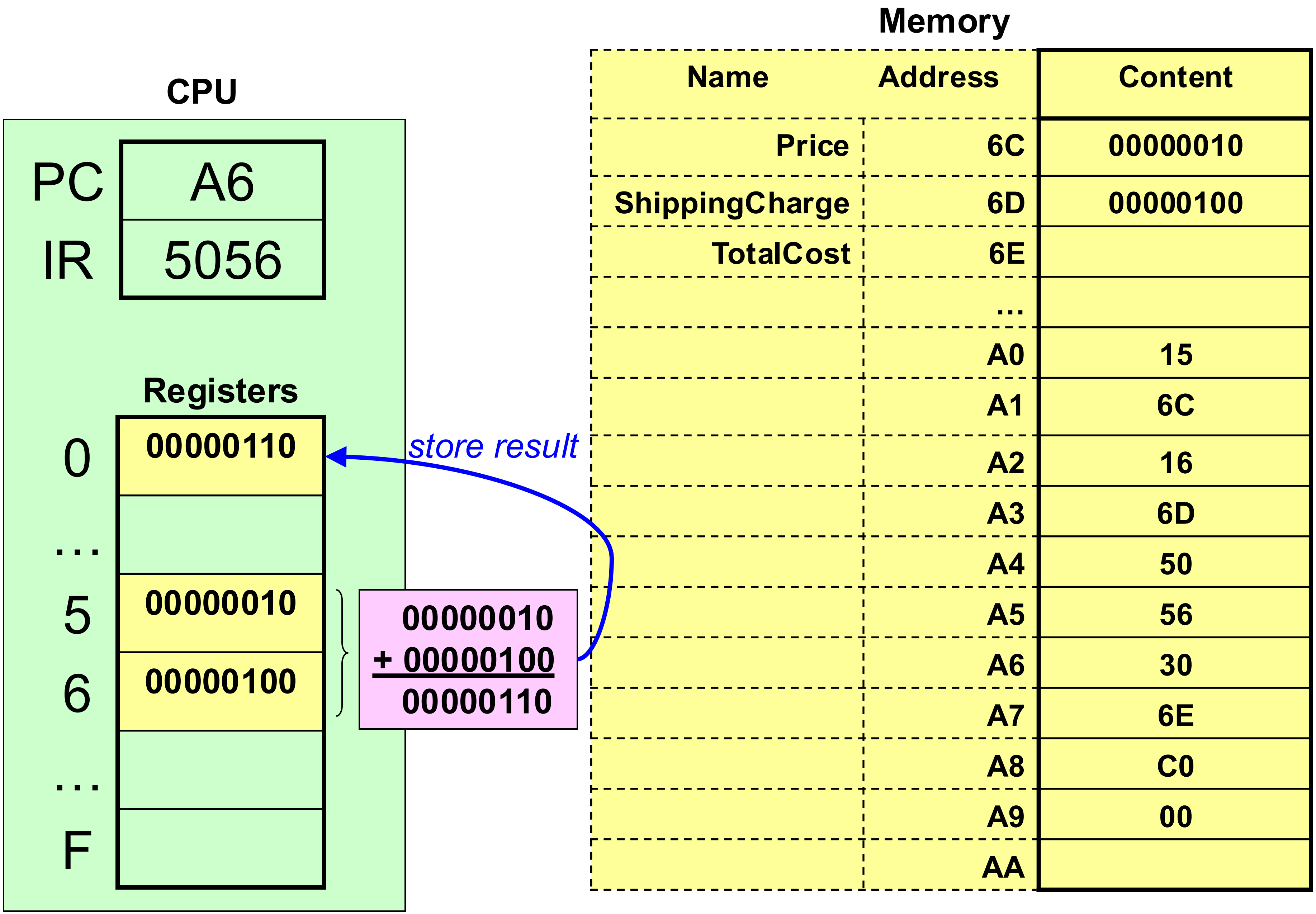
Registers

0	
…	
5	00000010
6	00000100
…	
F	

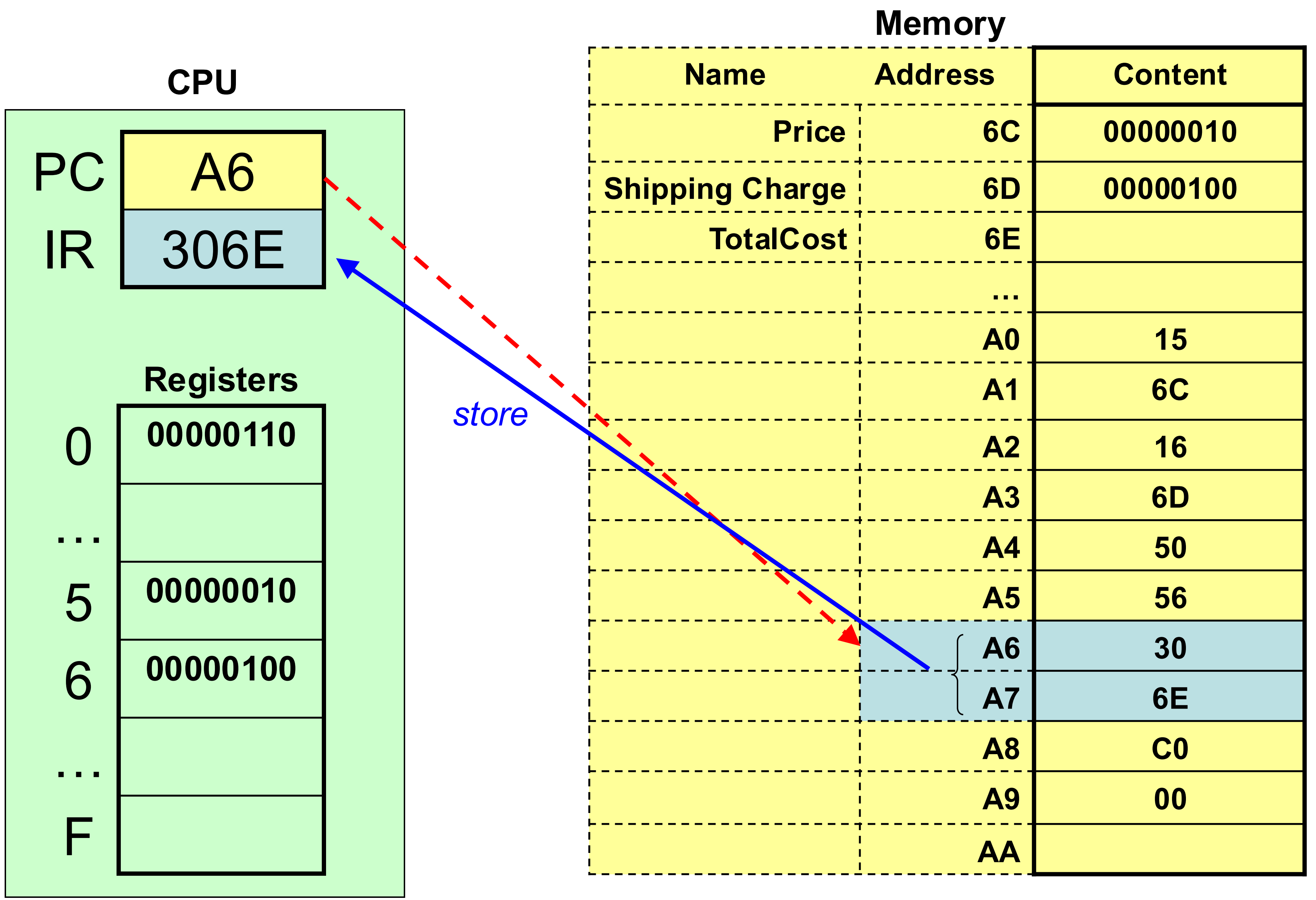
Memory

Name	Address	Content
Price	6C	00000010
ShippingCharge	6D	00000100
TotalCost	6E	
	…	
	A0	15
	A1	6C
	A2	16
	A3	6D
	A4	50
	A5	56
	A6	30
	A7	6E
	A8	C0
	A9	00
	AA	

c) **Execute** – perform addition and store



a) **Fetch -** instruction at A6 and A7 – 30, 6E and store it in IR; increment PC to A8



a) **Fetch -** instruction at A6 and A7 – 30, 6E and store it in IR; increment PC to A8

CPU

PC	A8
IR	306E

Registers

0	00000110
…	
5	00000010
6	00000100
…	
F	

Memory

Name	Address	Content
Price	6C	00000010
Shipping Charge	6D	00000100
TotalCost	6E	
	…	
	A0	15
	A1	6C
	A2	16
	A3	6D
	A4	50
	A5	56
	A6	30
	A7	6E
	A8	C0
	A9	00
	AA	

b) **Decode** – store value in register 0 at **TotalCost** (memory location 6E)

CPU

PC	A8
IR	306E

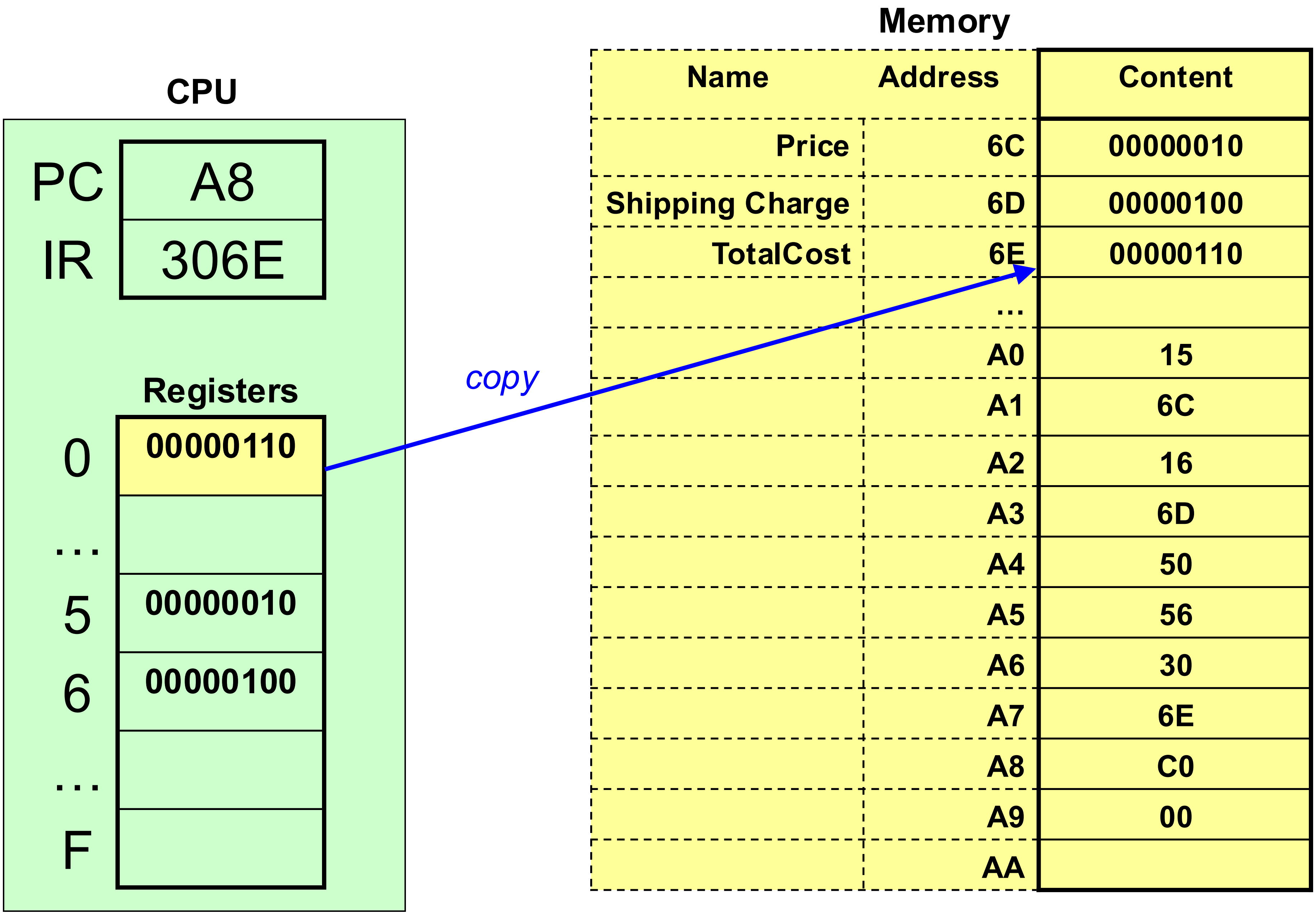
Registers

0	00000110
…	
5	00000010
6	00000100
…	
F	

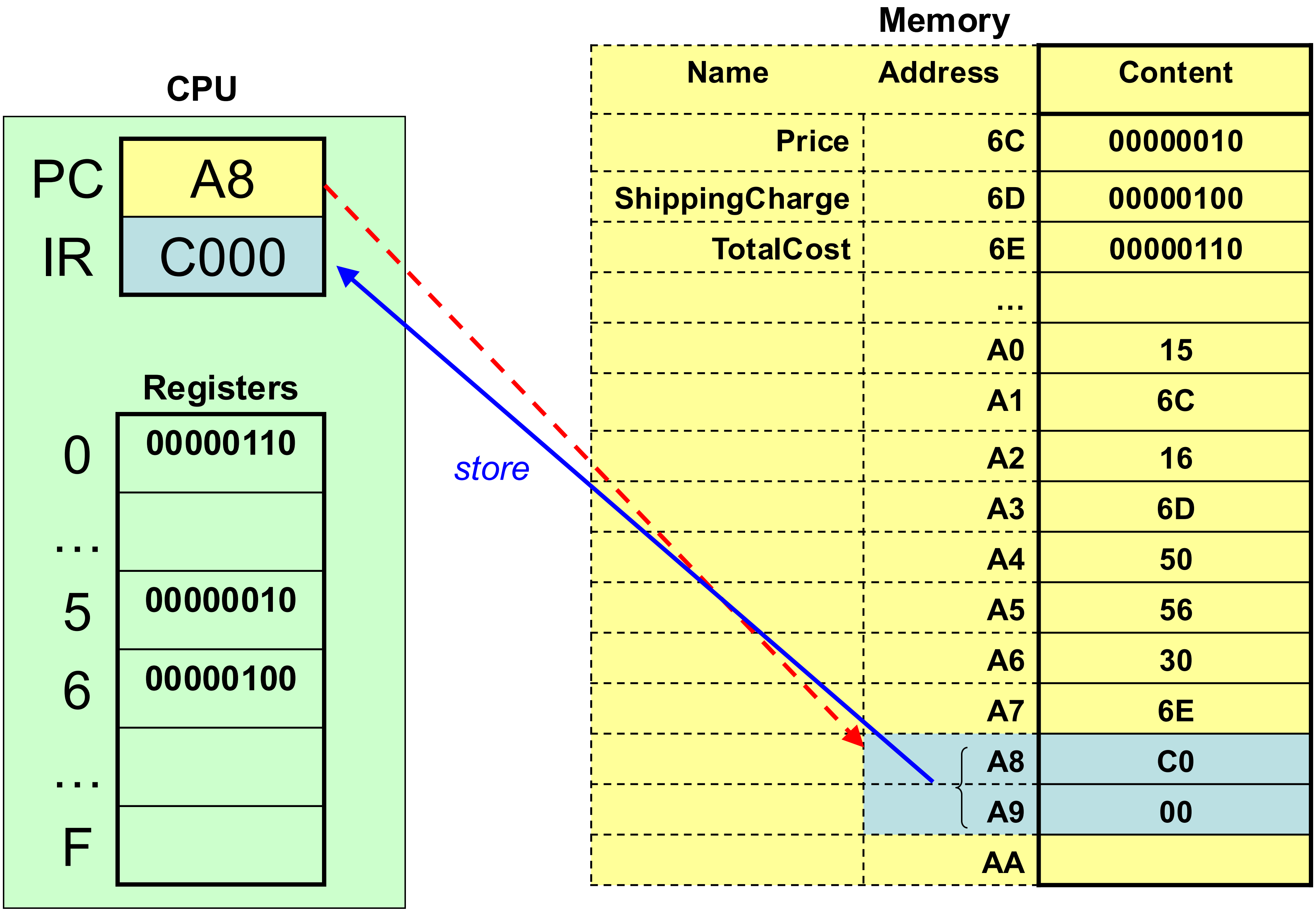
Memory

Name	Address	Content
Price	6C	00000010
Shipping Charge	6D	00000100
TotalCost	6E	
	…	
	A0	15
	A1	6C
	A2	16
	A3	6D
	A4	50
	A5	56
	A6	30
	A7	6E
	A8	C0
	A9	00
	AA	

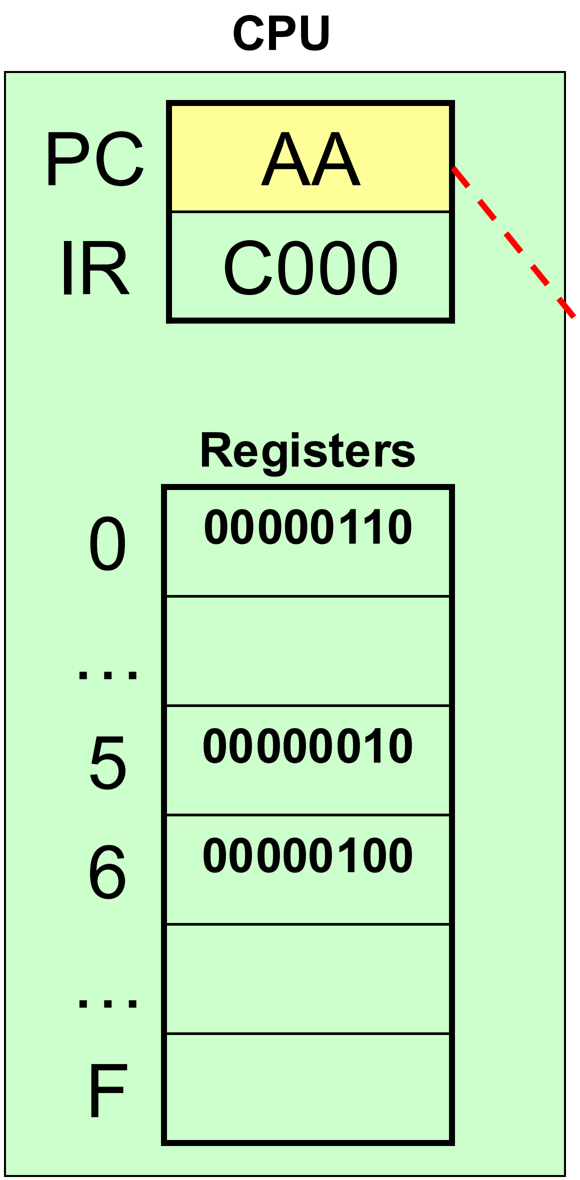
c) **Execute** – copy value from register 0 to memory location 6E



a) **Fetch -** instruction at A8 and A9 – C0, 00 and store it in IR; increment PC to AA



a) **Fetch -** instruction at A8 and A9 – C0, 00 and store it in IR; increment PC to AA



Memory

Name	Address	Content
Price	6C	00000010
ShippingCharge	6D	00000100
TotalCost	6E	00000110
	…	
	A0	15
	A1	6C
	A2	16
	A3	6D
	A4	50
	A5	56
	A6	30
	A7	6E
	A8	C0
	A9	00
	AA	

b) **Decode** – halt

CPU

PC	AA
IR	C000

Registers

0	00000110
…	
5	00000010
6	00000100
…	
F	

Memory

Name	Address	Content
Price	6C	00000010
ShippingCharge	6D	00000100
TotalCost	6E	00000110
	…	
	A0	15
	A1	6C
	A2	16
	A3	6D
	A4	50
	A5	56
	A6	30
	A7	6E
	A8	C0
	A9	00
	AA	

c) **Execute** – stop

CPU

PC	AA
IR	C000

Registers

0	00000110
…	
5	00000010
6	00000100
…	
F	

Memory

Name	Address	Content
Price	6C	00000010
ShippingCharge	6D	00000100
TotalCost	6E	00000110
	…	
	A0	15
	A1	6C
	A2	16
	A3	6D
	A4	50
	A5	56
	A6	30
	A7	6E
	A8	C0
	A9	00
	AA	